

Arya Mohammadi

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EDUCATION

University of California - San Diego

B.S. in Mathematics and Computer Science, GPA: 3.75/4.00

Expected June 2027

Dean's List (4x)

Coursework: Data Structures, Algorithms, Software Construction, Systems Programming & Software Tools, Assembly Language, Discrete Mathematics

TECHNICAL SKILLS

Languages: Python, C++, C, Java, JavaScript, TypeScript, SQL

Frameworks & Libraries: Next.js, React.js, Node.js, Express.js, Mongoose, Jest, Supertest, GoogleTest, Playwright

Tools & Platforms: Git, Linux, MongoDB, Vercel, Railway, Cloudinary, Twilio, CMake, gdb, Valgrind

Concepts: Systems programming, debugging, REST API design, authentication, unit testing, CI/CD

EXPERIENCE

Computer Science Tutor

September 2024 – June 2025

University of California, Riverside

- Led 8–12 weekly sessions for 200+ students across 3 quarters, teaching recursion, sorting, graphs, and complexity analysis.
- Debugged student C++ codebases, identifying issues with memory management, pointer logic, and off-by-one errors.
- Created 20+ practice problems adopted by staff for exam preparation, covering recursion, Big-O analysis, & graph traversal.
- Taught Git fundamentals (branching, commits, merges) to students new to version control.

Hardware Intern

June 2021 – September 2023

ProCustom Group

- Assembled 250+ rugged laptop systems for Lockheed Martin and Raytheon with 100% on-time delivery.
- Fabricated 160+ custom cables and applied EMI shielding, achieving 100% military standard compliance.
- Documented 50+ production tasks for quality control, contributing to AS9100D certification maintenance.

PROJECTS

DormDuos.com | *React.js, Node.js, Express.js, MongoDB, JWT, Jest, Vercel, Railway*

- Built a housing platform for UCR students to find & post off-campus listings; React frontend, Express API, JWT auth, Cloudinary uploads.
- Added input sanitization to prevent NoSQL injection, bcrypt password hashing, and protected routes for secure access.
- Optimized search across 10k+ listings with compound indexes and pagination, and wrote tests with Jest & Supertest to achieve 100% branch coverage.

CodeLens | *Next.js, TypeScript, OpenAI API, Zustand, Tailwind CSS, Vercel*

- Built a code review tool that analyzes snippets across 8 languages using GPT-4, showing bugs, optimizations, and time/space complexity.
- Wrote structured prompts with strict JSON schemas for consistent API responses, and added validation to handle empty or malformed outputs.
- Managed async state with Zustand and built syntax highlighting, skeleton loaders, and staggered animations for the UI.

AryaElectric.com | *Next.js, Node.js, Twilio, Nodemailer, Jest, Playwright*

- Built a production website for an electrical contractor to streamline quote requests and follow-ups.
- Implemented async API routes for quote submissions, SMS webhooks, and automated follow-ups.
- Added server-side caching for Google Reviews API, reducing response latency by 90%.
- Wrote 90+ tests with mocked Twilio/Nodemailer; validated end-to-end flows with Playwright.

Battlestar RPG | *C++, STL, GoogleTest, CMake*

- Built 3.4k-line RPG engine with 23 classes + 2.3k-line test cases over 3 months in a team of 5.
- Implemented MaxHeap-based combat scheduler for $O(n \log n)$ speed-based turn ordering.
- Designed polymorphic item system with clone pattern and template-based save/load serialization.

Chat Server | *C, HTTP, Sockets, Linux*

- Built a multi-client chat server in C using POSIX sockets with manual HTTP parsing and routing for a messaging system with posts and reactions.
- Created in-memory data structures with size limits, timestamps, and UTF-8 validation to handle bad input and prevent buffer overflows.
- Tested under concurrent connections on Linux, checking memory usage and verifying correct request handling.